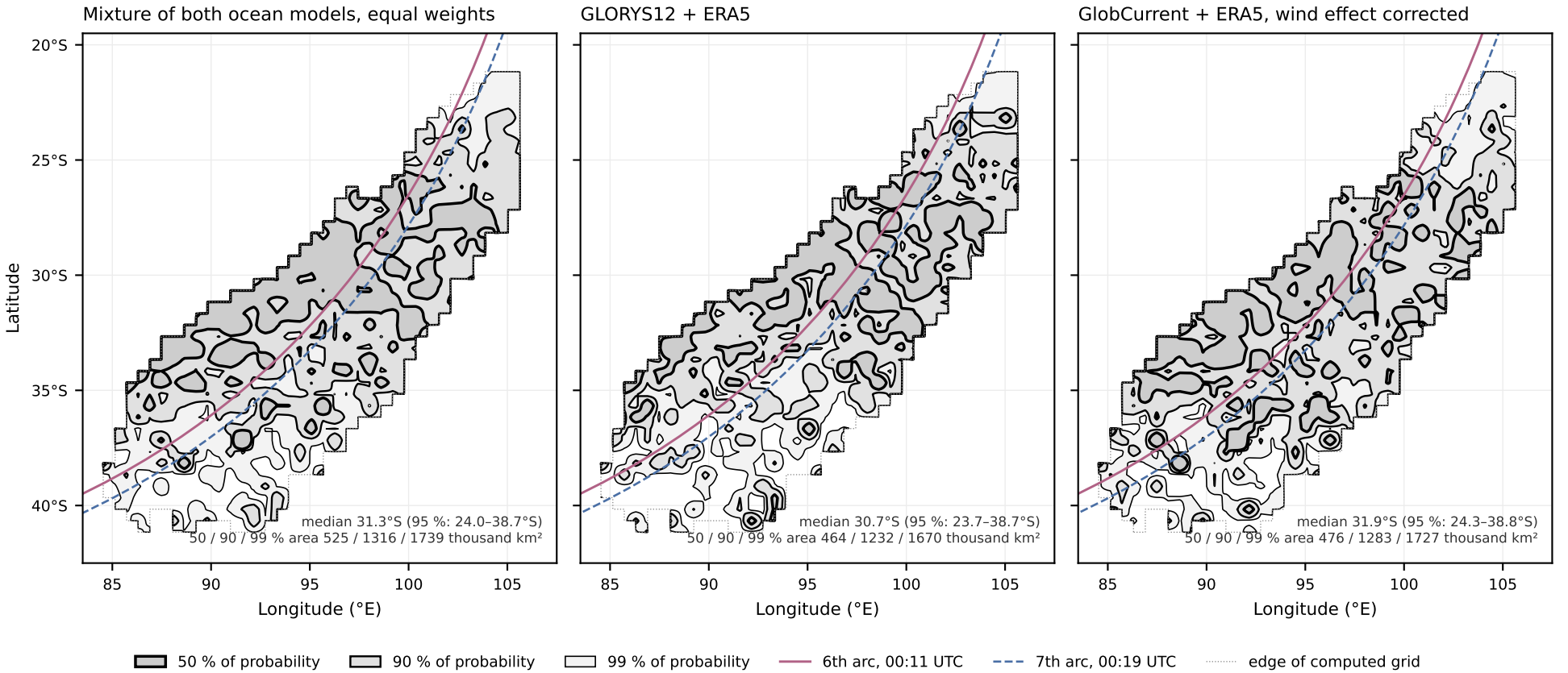
Impact location from the recovered debris alone: posterior PDF, 50 / 90 / 99 % regions



What the chart shows. Each map shows where the aircraft hit the sea, using only the drift of the nine identified debris items. The darker the grey, the more probable the area. The left map averages the two ocean models with equal weight. The other maps show each model alone. The maps do not use the satellite data or the flight path, except that the calculation is limited to the grid (dotted line), which is placed where the flight model puts impacts. Values in the far south are noisy. The results are interim and are not evidence of the impact location.

Technical. Debris-drift log-likelihood, nine stringent finds, recovery bandwidth 50 km; run A surfaces on 710 nodes at 30 NM (GLORYS12V1 0.494 m: production 367 + extension 343; GlobCurrent my-p1d 0 m, windage product-relative, c_wind offset U(-0.75 %, -0.60 %) of U10, audit F1), each + ERA5 10 m wind; bilinear interpolation in likelihood (interpolate.rs) on a 0.02° grid; uniform prior per unit area over the support scored by both models (6 GLORYS12 and 1 GlobCurrent nodes unresolved at 41.2-39.2°S, not computed). Mixture = 0.5 p_GLORYS12 + 0.5 p_GlobCurrent (equal prior model weight, Pete 03:10Z). Posterior model weights from the debris evidence over this support: GLORYS12 0.62, GlobCurrent 0.38; evidence-weighted mixture median 31.1°S. Split-half Monte Carlo noise per node 1.61 (GLORYS12), 1.17 (GlobCurrent) log-likelihood units; NOT CONVERGED in the south. HPD levels on the unsmoothed density. Arcs: core run D reference_arcs (m0011, m0019a). Not combined with core/end-of-flight posteriors, searches, hydroacoustics or Pléiades.